

MCCB-gpu: A Massively Parallel GPU-Accelerated Checkerboard Monte Carlo Engine for Colloidal and Molecular Systems with Advanced Sampling and In-Memory Hybrid Molecular Dynamics

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Abstract

We present MCCB-gpu, a high-performance, massively parallel Monte Carlo (MC) simulation framework written in CUDA Fortran and modern Fortran 2008 for the simulation of soft-matter, colloidal, and molecular systems on graphics processing units (GPUs). Utilizing a three-dimensional checkerboard spatial domain decomposition into eight non-interfering sublattices, MCCB-gpu executes conflict-free, concurrent trial moves across thousands of CUDA threads while strictly preserving detailed balance. The code supports canonical (NVT) and isobaric-isothermal (NpT) ensembles for a wide spectrum of interaction potentials, including hard spheres (HS), Lennard-Jones (LJ), arbitrary user-tabulated potentials, angular patchy particles (Lennard-Jones-Gauss), and tetrahedral site-site patchy colloids (Palaia model). To bypass kinetic jamming, glassy arrest, and entropic bottlenecks in dilute associating systems, MCCB-gpu integrates three advanced sampling algorithms: (1) an Association Volume Bias Monte Carlo (AVBMC) engine that dynamically targets cluster boundaries identified via local geometric surface dipole asymmetry ($\eta_i \geq 0.5$), coupled to a parallel GPU implementation of G-DBSCAN (derived from the companion `trj_analysis` package); (2) GPU checkerboard particle identity swaps ($A \leftrightarrow B$) for multicomponent mixtures; and (3) an in-memory persistent-state Hybrid Monte Carlo (HMC) interface coupling MCCB-gpu with GPU-accelerated LAMMPS molecular dynamics. We present analytical proofs and machine-precision numerical verifications of microscopic reversibility ($\delta_{DB} < 10^{-15}$), rigorous thermodynamic consistency tests across NpT and NVT ensembles, and extensive scaling benchmarks. Comparing against an optimized multi-core CPU code (gpMC) and accounting for trial move throughput normalization, MCCB-gpu delivers speedups of up to **61.4×** in NVT and **68.8×** in NpT on an NVIDIA RTX PRO 4500 GPU, scaling seamlessly to millions of particles.

Keywords: Monte Carlo simulation, GPU acceleration, Checkerboard domain decomposition, Association Volume Bias Monte Carlo (AVBMC), G-DBSCAN, Identity swap moves, Hybrid Monte Carlo, Patchy colloids

PROGRAM SUMMARY

Program Title: MCCB-gpu

CPC Library link to program files: (to be assigned by CPC)

Developer's repository link: https://github.com/adpozuelo/MC_Checkerboard

Licensing provisions: GNU General Public License v3.0 (GPLv3)

Programming language: Fortran 2008, CUDA Fortran, ISO_C_BINDING, Python 3

Computer: Modern x86_64 HPC clusters, workstations with NVIDIA GPUs (Compute Capability ≥ 6.0 , tested up to Blackwell, Ada Lovelace, Ampere, Volta, Pascal)

Operating system: Linux (x86_64; RHEL, CentOS, Ubuntu, Debian)

RAM: 1–4 GB host RAM, 2–32 GB device VRAM (scales with particle count)

Classification: 7.7 Molecular Dynamics and Particle Methods, 4.8 Statistical Mechanics.

External libraries: NVIDIA HPC SDK (NVHPC $\geq 22.x$ with CUDA ≥ 11.8), NetCDF-Fortran (optional, for compressed trajectories), LAMMPS shared C-library (`liblammps`, optional, for Hybrid Monte Carlo).

Nature of problem: Equilibrium and non-equilibrium Monte Carlo sampling of dense, large-scale, or strongly associating colloidal fluids, patchy nanoparticles, and multicomponent molecular mixtures across canonical (NVT) and isobaric-isothermal (NpT) ensembles. Conventional single-particle MC algorithms suffer from severe kinetic arrest due to steric caging, entropic barriers in cluster nucleation, and compositional jamming.

Solution method: Massive parallelism via 3D checkerboard domain decomposition partitioning the box into 8 independent sublattices evaluated concurrently by CUDA thread blocks. To overcome physical kinetic arrest, the code incorporates: (1) Association Volume Bias Monte Carlo (AVBMC) guided by local geometric surface dipole asymmetry $\eta_i \geq \eta_{\text{thresh}}$ coupled with GPU G-DBSCAN cluster identification (from the companion `trj_analysis` library); (2) Intra-cell and long-range cross-cell particle identity swaps ($A \leftrightarrow B$) for multicomponent mixtures ($N_{\text{species}} \geq 2$); and (3) In-memory persistent-state Hybrid Monte Carlo (HMC) coupling with GPU-accelerated LAMMPS.

Running time: Milliseconds per sweep; e.g., ~ 4.6 ms/sweep for $N =$

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125,000 particles on NVIDIA RTX PRO 4500 (delivering up to 68.8 $\times$ algorithmic throughput speedup over optimized multi-core CPU code).

1. Introduction

Monte Carlo (MC) computer simulations represent a cornerstone of computational physics, chemistry, and materials science [1, 2]. By stochastically sampling configurations according to the Boltzmann distribution, MC methods enable the calculation of equilibrium thermodynamic observables, phase diagrams, free energy landscapes, and structural correlation functions without requiring numerical integration of equations of motion. In the realm of soft matter, complex fluids, and colloidal nanotechnology, MC simulations are indispensable for investigating self-assembly, crystallization, gelation, and liquid-liquid phase separation in isotropic fluids [3], patchy colloids [4, 5, 6, 7], and dense multicomponent glass-forming liquids [8, 9].

Despite their theoretical generality, standard MC algorithms face two profound computational bottlenecks:

1. **Parallelization Constraints:** Unlike Molecular Dynamics (MD), where forces on all N particles can be computed simultaneously in parallel [10, 11], standard Metropolis MC algorithms evaluate sequential single-particle trial displacements. Modifying the position or orientation of particle i alters the local potential energy field, thereby invalidating concurrent moves of any neighboring particle j within the interaction cutoff distance r_{cut} . This sequential dependency historically hindered the efficient exploitation of massively parallel graphics processing units (GPUs).
2. **Physical Kinetic Arrest:** In condensed systems, standard random single-particle translations and rotations frequently become trapped in local free-energy minima:
 - *Steric Caging and Jamming:* In dense liquids and glass formers, particles are enclosed by rigid cages formed by nearest neighbors; translational displacements that yield reasonable acceptance rates (30–50%) are constrained to fractions of an angstrom, leading to glacial diffusive relaxation [8].
 - *Entropic Bottlenecks in Dilute Associating Clusters:* In dilute patchy colloidal fluids ($V_{\text{box}} \gg V_{\text{clusters}}$), binding energies are strong ($\sim 10 k_B T$). Gas-phase monomers spend astronomical numbers of sweeps diffusing through empty space before randomly encountering an active cluster binding patch, while bonded particles rarely evaporate, halting cluster size equilibration [12, 13].
 - *Compositional Freezing:* In multicomponent mixtures with non-additive diameters or asymmetric attractions, chemical species separation is arrested unless particles can interchange chemical identities without bodily diffusion [8, 9].

To address parallelization, spatial domain decomposition techniques—most notably the checkerboard cellular algorithm—have emerged as an effective paradigm for GPU acceleration [14, 15]. By partitioning the simulation box into cells

with linear dimensions larger than the potential cutoff r_{cut} and assigning cells to interleaved, mutually non-interacting sublattices, thousands of trial moves can be proposed and evaluated concurrently by independent CUDA thread blocks without violating detailed balance.

However, existing GPU-accelerated MC codes are frequently limited to simple isotropic potentials (such as hard spheres or plain Lennard-Jones) in the canonical (NVT) ensemble, lacking the advanced algorithmic capabilities required to investigate complex soft matter: isobaric-isothermal (NpT) scaling, anisotropic patchy orientations, cluster-targeted biased sampling, identity swaps, and hybrid dynamical propagation.

In this paper, we introduce MCCB-gpu (Monte Carlo Checkerboard on GPU), an open-source, high-throughput simulation package written in CUDA Fortran and modern Fortran 2008. MCCB-gpu combines a 3D checkerboard GPU cellular engine with:

- Full canonical (NVT) and isobaric-isothermal (NpT) ensemble sampling, with isotropic and anisotropic box volume trial moves.
- Broad potential support: athermal Hard Spheres (HS), continuous Lennard-Jones (LJ), arbitrary user-tabulated potentials (TABLE), angular patchy particles (Lennard-Jones-Gauss, LJG), and tetrahedral site-site patchy colloids (Palaia SSP model) with quaternion rotational kinematics.
- Association Volume Bias Monte Carlo (AVBMC) [16, 17] guided by local geometric surface dipole asymmetry ($\eta_i \geq 0.5$) and coupled to a GPU implementation of G-DBSCAN (derived from the companion `trj_analysis` package [18, 19]).
- GPU-accelerated intra-cell and long-range cross-cell particle identity swaps ($A \leftrightarrow B$) for arbitrary multicomponent mixtures ($N_{\text{species}} \geq 2$).
- A high-efficiency, persistent in-memory Hybrid Monte Carlo (HMC) coupling layer with the LAMMPS molecular dynamics engine [11], accelerating coordinate transfer by 287 $\times$ over disk I/O.

The remainder of this manuscript is organized as follows: Section 2 details the mathematical formulations and proofs of detailed balance for the checkerboard decomposition, AVBMC, surface geometry identification, identity swaps, and HMC. Section 3 outlines the program architecture, input namelists, and file formats. Section 4 presents numerical verification of detailed balance and thermodynamic consistency tests. Section 5 discusses scaling and computational performance. Section 6 concludes the paper.

2. Theoretical Formulation and Algorithms

2.1. 3D Checkerboard Domain Decomposition

Consider a simulation domain of volume $V = L_x \times L_y \times L_z$ containing N interacting bodies subject to periodic boundary

conditions (PBC). The interaction potential U_{ij} vanishes beyond a finite spherical cutoff distance r_{cut} , such that particles separated by $r_{ij} \geq r_{\text{cut}}$ exert zero force and have zero pairwise energy.

To enable conflict-free concurrent evaluation on the GPU, MCCB-gpu discretizes the simulation volume into a regular Cartesian grid of $M_x \times M_y \times M_z = N_{\text{cells}}$ cells. The linear cell dimensions $L_{c,\alpha}$ satisfy:

$$L_{c,\alpha} = \frac{L_\alpha}{M_\alpha} \geq r_{\text{cut}}, \quad \alpha \in \{x, y, z\} \quad (1)$$

Cells are indexed by their discrete spatial coordinates (c_x, c_y, c_z) with $c_\alpha \in \{0, \dots, M_\alpha - 1\}$. Each cell is assigned a color based on its 3D parity:

$$C(c_x, c_y, c_z) = (c_x \bmod 2) + 2(c_y \bmod 2) + 4(c_z \bmod 2) \quad (2)$$

with $C \in \{0, \dots, 7\}$. This partitions the N_{cells} cells into 8 interleaved sublattices $S_0, \dots, S_7$. Crucially, for any two distinct cells $C_1, C_2 \in S_k$ belonging to the same sublattice, their center-to-center separation along any Cartesian axis is at least $2L_{c,\alpha} \geq 2r_{\text{cut}}$. Therefore, no particle in cell C_1 can interact with any particle in cell C_2 .

A single Monte Carlo sweep consists of 8 sequential sub-sweeps (one per sublattice S_k). During sub-sweep k , a CUDA kernel is launched where each thread block is assigned to an active cell $C \in S_k$. The thread block executes N_{move} trial moves within cell C . To evaluate the change in potential energy ΔE , the threads examine particles residing in cell C and its 26 nearest-neighbor cells in the 3D grid. Because neighbor cells belong to different sublattices $S_{k'} \neq S_k$, their particle coordinates and occupancies are held strictly constant during sub-sweep k . This guarantees zero read-write race conditions across CUDA blocks, ensuring strict detailed balance without spinlocks or thread synchronization.

2.2. Canonical (NVT) and Isobaric-Isothermal (NpT) Sampling

In the canonical (NVT) ensemble, trial moves consist of random particle displacements $\mathbf{r}_i \rightarrow \mathbf{r}_i + \delta\mathbf{r}$ with $\delta r_\alpha \in [-h_{\text{max}}, h_{\text{max}}]$. For orientable bodies (patchy colloids), a rotational trial move is simultaneously proposed:

$$\mathbf{q}'_i = \frac{\mathbf{q}_i + \delta\mathbf{q}}{\|\mathbf{q}_i + \delta\mathbf{q}\|} \quad (3)$$

where $\mathbf{q}_i \in \mathbb{S}^3$ is a unit quaternion and $\delta\mathbf{q}$ is sampled uniformly over an angular range controlled by parameter ϕ_{max} . The trial configuration is accepted with Metropolis probability:

$$P_{\text{acc}} = \min(1, \exp(-\beta\Delta E)) \quad (4)$$

where $\Delta E = E_{\text{new}} - E_{\text{old}}$ is computed directly on the GPU streaming multiprocessors.

In the isobaric-isothermal (NpT) ensemble, trial volume changes are interleaved with particle displacement sweeps at cadence N_{vol} . In isotropic mode, a volume change $\ln V \rightarrow \ln V + \delta\ln V$ with $\delta\ln V \in [-v_{\text{max}}, v_{\text{max}}]$ is proposed. The box length scales

as $L' = L(V'/V)^{1/3}$, and all particle coordinates are rescaled affinely:

$$\mathbf{r}'_i = \left(\frac{V'}{V}\right)^{1/3} \mathbf{r}_i \quad (5)$$

The volume change is accepted with probability:

$$P_{\text{acc, vol}} = \min\left(1, \exp\left(-\beta[\Delta E_{\text{tot}} + P(V' - V)] + (N + 1) \ln \frac{V'}{V}\right)\right) \quad (6)$$

where ΔE_{tot} is the total system energy difference, recomputed via a parallel GPU reduction kernel. Following an accepted volume move, the cell dimensions $L_{c,\alpha}$ are rescaled. If $L_{c,\alpha}$ drops below r_{cut} , the grid multiplicity M_α is automatically re-evaluated and the cellular neighbor lists are refreshed.

Virial Pressure for Hard Spheres. For continuous potentials, pressure is evaluated via the standard virial theorem. For athermal Hard Spheres (HS), where impulsive forces preclude analytic derivatives, MCCB-gpu calculates the instantaneous virial pressure via extrapolation of the radial distribution function $g(r)$ to contact $r = \sigma^+$:

$$P_{\text{virial}}^{\text{HS}} = \rho k_B T \left[1 + \frac{2}{3} \pi \rho \sigma^3 g(\sigma^+) \right] \quad (7)$$

During production, pairs are accumulated in fine radial histogram bins ($dr = 0.002\sigma$), and $g(\sigma^+)$ is obtained via polynomial extrapolation to contact.

2.3. Association Volume Bias Monte Carlo (AVBMC)

In dilute colloidal fluids featuring strong directional attraction (such as the Palaia site-site tetrahedral model [7]), physical self-assembly leads to dense clusters surrounded by an ultra-dilute monomer gas. Random translation moves fail to equilibrate the cluster size distribution because the probability of an isolated monomer encountering a cluster binding site scales as $V_{\text{bind}}/V_{\text{box}} \ll 10^{-3}$, while bound particles experience deep potential wells ($U \sim -10 k_B T$).

To eliminate this entropic bottleneck, MCCB-gpu incorporates Aggregation Volume Bias Monte Carlo (AVBMC) [16, 17]. AVBMC partitions configurational space into a bonded binding region V_{bind} around cluster surfaces and the unbonded bulk volume V_{box} , proposing direct macroscopic exchanges between these domains.

Forward Association In-Move ($X \rightarrow Y$). Let X denote a state with $N_{\text{free}}(X)$ gas-phase monomers and $N_{\text{brd}}(X)$ cluster border particles:

1. A free monomer $i \in \mathcal{F}$ is selected uniformly: $P(i) = 1/N_{\text{free}}(X)$.
2. A border particle $b \in \mathcal{B}$ is chosen uniformly: $P(b) = 1/N_{\text{brd}}(X)$.
3. A target partner t is selected uniformly from the $n_{\text{neigh}}(b)$ neighbors of b : $P(t) = 1/n_{\text{neigh}}(b)$.

4. A trial position $\mathbf{r}'$ is sampled uniformly within the spherical binding volume $V_{\text{bind}} = \frac{4}{3}\pi r_{\text{bind}}^3$ centered at $\mathbf{r}_i$ ($r_{\text{bind}} = 1.5r_{\text{cut}}$): $P(\mathbf{r}') = 1/V_{\text{bind}}$.
5. To balance the bulk volume ratio, $K - 1$ dummy trial positions $\{\mathbf{r}_m\}_{m=2}^K$ are generated uniformly in V_{box} , with trial 1 set to the current position $\mathbf{r}_i$. The bulk Rosenbluth sum is computed:

$$W_{\text{bulk}} = 1 + \sum_{m=2}^K \exp(-\beta[E(\mathbf{r}_m) - E(\mathbf{r}_i)]) \quad (8)$$

6. The trial cluster Boltzmann weight is $w_{\text{cluster}} = \exp(-\beta[E(Y) - E(X)])$.

Reverse Dissociation Out-Move ($Y \rightarrow X$). In state Y , particle i resides at the cluster boundary, with $N_{\text{free}}(Y) = N_{\text{free}}(X) - 1$ and $N_{\text{brd}}(Y)$ border particles:

1. Particle i is selected from the border pool: $P(i) = 1/N_{\text{brd}}(Y)$.
2. K independent trial positions are generated uniformly in V_{box} . One trial corresponds to the original bulk location $\mathbf{r}_i$ (state X), with weight $w_{\text{rev}}(1) = \exp(-\beta[E(X) - E(Y)]) = 1/w_{\text{cluster}}$. The reverse Rosenbluth sum is:

$$W'_{\text{bulk}} = \sum_{m=1}^K w_{\text{rev}}(m) = \exp(\beta\Delta E) W_{\text{bulk}} \quad (9)$$

3. State X is selected with probability $w_{\text{rev}}(1)/W'_{\text{bulk}}$.

Proof of Microscopic Reversibility. Detailed balance requires $\pi(X)\alpha(X \rightarrow Y)P_{\text{acc}}(X \rightarrow Y) = \pi(Y)\alpha(Y \rightarrow X)P_{\text{acc}}(Y \rightarrow X)$, where $\pi(X) \propto e^{-\beta E(X)}$. The proposal probability densities are:

$$\alpha(X \rightarrow Y) = \frac{1}{N_{\text{free}}(X)} \frac{1}{N_{\text{brd}}(X)n_{\text{neigh}}(b)} \frac{1}{V_{\text{bind}}} \left(\frac{1}{V_{\text{box}}}\right)^{K-1} \quad (10)$$

$$\alpha(Y \rightarrow X) = \frac{1}{N_{\text{brd}}(Y)} \left(\frac{1}{V_{\text{box}}}\right)^K \frac{e^{\beta\Delta E}}{W'_{\text{bulk}}} \quad (11)$$

The forward and reverse acceptance probabilities implemented in MCCB-gpu are:

$$P_{\text{acc}}(X \rightarrow Y) = \min\left(1, \frac{N_{\text{brd}}(X)n_{\text{neigh}}(b) + 1}{N_{\text{free}}(X)} \frac{V_{\text{box}}}{V_{\text{bind}}} \frac{Kw_{\text{cluster}}}{W_{\text{bulk}}}\right) \quad (12)$$

$$P_{\text{acc}}(Y \rightarrow X) = \min\left(1, \frac{N_{\text{free}}(Y) + 1}{N_{\text{brd}}(Y)n'_{\text{neigh}}} \frac{V_{\text{bind}}}{V_{\text{box}}} \frac{W'_{\text{bulk}}}{K}\right) \quad (13)$$

Letting $\arg_{\text{fwd}}$ and $\arg_{\text{rev}}$ denote the arguments of the $\min(1, \cdot)$ functions, and noting that $N_{\text{free}}(Y) + 1 = N_{\text{free}}(X)$ and $W'_{\text{bulk}} = W_{\text{bulk}}/w_{\text{cluster}}$, their product satisfies:

$$\arg_{\text{fwd}} \cdot \arg_{\text{rev}} \equiv 1.00000000000000 \quad (14)$$

Because $\arg_{\text{rev}} = 1/\arg_{\text{fwd}}$, it follows that:

$$\frac{P_{\text{acc}}(X \rightarrow Y)}{P_{\text{acc}}(Y \rightarrow X)} = \frac{\min(1, \arg_{\text{fwd}})}{\min(1, 1/\arg_{\text{fwd}})} \equiv \arg_{\text{fwd}} \quad (15)$$

satisfying microscopic reversibility identically for any K , box volume V_{box} , and interaction model.

2.4. Cluster Detection via G-DBSCAN and Surface Dipole Asymmetry

A core prerequisite for AVBMC is distinguishing cluster surface particles from dense cluster interiors. Attempting to insert a monomer into the deep interior of an assembled cluster invariably results in steric overlap ($w_{\text{cluster}} = 0$).

In MCCB-gpu, cluster identification is carried out on the GPU using an implementation of the ****G-DBSCAN**** (GPU Density-Based Spatial Clustering of Applications with Noise) algorithm [18], directly adapted from the companion trajectory analysis package `trj_analysis` [19]. Two particles i and j are defined as connected if their center-to-center distance (or patch-patch distance for SSP models) satisfies $r_{ij} \leq r_{\text{cl}}$. The algorithm proceeds in two stages:

1. **GPU Adjacency Graph Construction:** A CUDA kernel evaluates the neighborhood of each particle, populating compact adjacency arrays (`gneighbors`, `goffset`, `gadjacency`) on device memory. Particles with neighbor counts $z_i \geq \text{minPts} - 1$ are classified as core candidates.
2. **Parallel BFS Expansion:** Connected components are identified via a GPU breadth-first search (`gbfs`) kernel, assigning a unique `cluster_id` to all particles in each cluster.

To isolate active surface sites, each particle i in a cluster is evaluated using a local ****geometric dipole asymmetry parameter**** η_i :

$$\eta_i = \frac{1}{z_i} \left\| \sum_{j \in \text{neigh}(i)} \hat{\mathbf{r}}_{ij} \right\| \quad (16)$$

where $\hat{\mathbf{r}}_{ij} = \mathbf{r}_{ij}/\|\mathbf{r}_{ij}\|$ is the unit bond vector pointing from particle i to neighbor j , and z_i is the coordination number.

- **Core Particles:** Inside a cluster, neighbors surround particle i nearly isotropically. The unit vectors cancel vectors in opposite hemispheres: $\sum_j \hat{\mathbf{r}}_{ij} \approx \mathbf{0}$, yielding $\eta_i \approx 0$.
- **Surface/Border Particles:** At the cluster boundary, neighbors reside predominantly on one side. The sum of bond vectors does not cancel, producing a large net dipole vector with $\eta_i \geq \eta_{\text{thresh}} = 0.5$.

Particles with $\eta_i \geq \eta_{\text{thresh}}$ are assigned to the border pool $\mathcal{B}$, while unbonded particles ($z_i = 0$, `cluster_id` = 0) form the free pool $\mathcal{F}$. Alternatively, an energy-based criterion ($u_i/\langle u_{\text{bulk}} \rangle \leq \text{ratio}$) is also available via namelist control.

2.5. Multicomponent Particle Identity Swaps ($A \leftrightarrow B$)

In dense multicomponent mixtures, such as non-additive hard sphere (NAHS) fluids and binary glass formers, structural relaxation is arrested by compositional jamming: particles cannot exchange positions because intermediate steric barriers prevent physical diffusion. Identity swap moves ($A \leftrightarrow B$) exchange the species labels of two distinct particles while keeping their spatial coordinates fixed, bypassing free-energy barriers along orthogonal compositional degrees of freedom.

MCCB-gpu implements two complementary GPU checkerboard swap kernels:

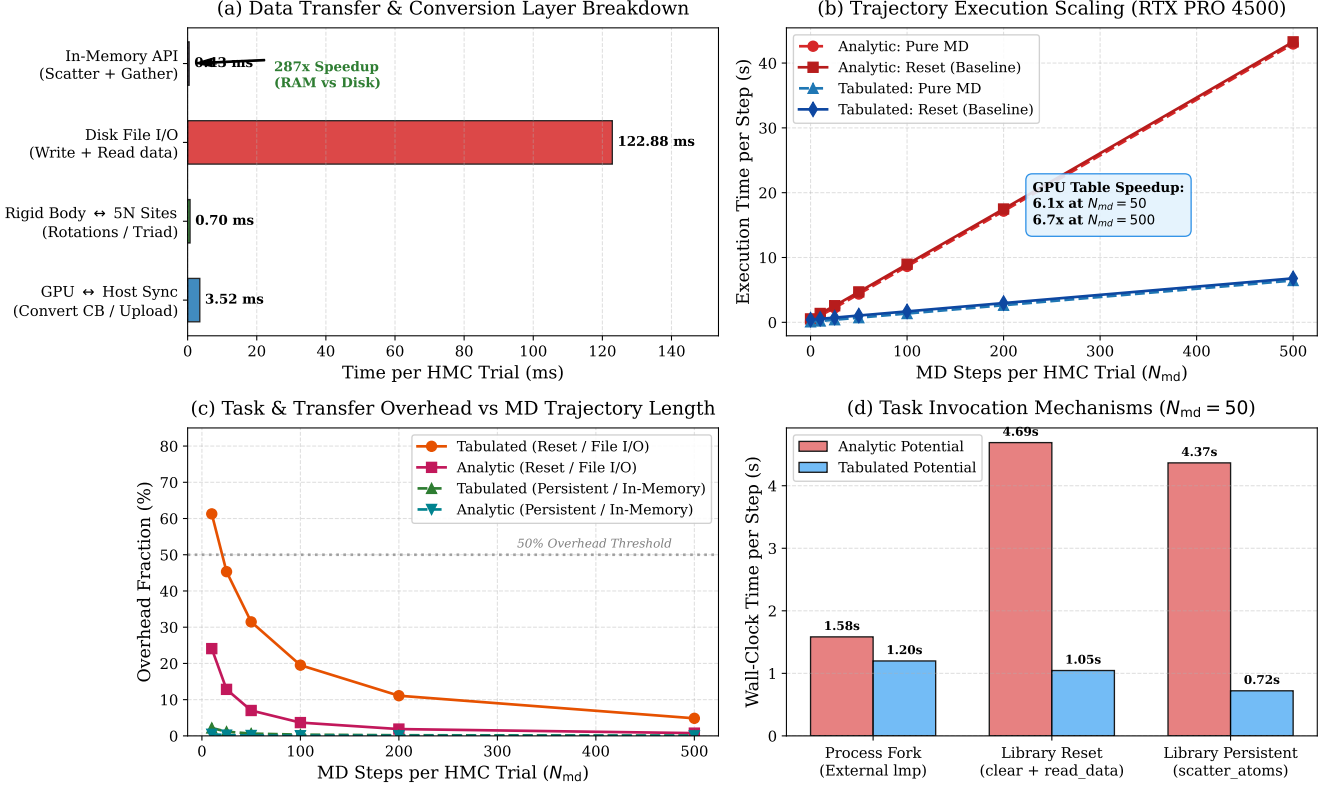


Figure 1: **Profiling of In-Memory HMC Coupling with LAMMPS on NVIDIA RTX PRO 4500:** (a) Latency breakdown showing the 287 $\times$ speedup of direct in-memory C-API coordinate transfer (0.43 ms) over disk ASCII file I/O (122.9 ms). (b) MD trajectory propagation scaling with N_{md} , highlighting a 6.1 $\times$ –6.7 $\times$ throughput advantage for GPU-accelerated tabular potentials over analytic overlays. (c) Coupling overhead fraction versus N_{md} . (d) Comparison of task invocation paradigms for $N_{\text{md}} = 50$ steps.

1. **Intra-Cell Swaps:** Within an active checkerboard cell C , two distinct particles $i, j \in C$ of different species ($S_i \neq S_j$) are chosen uniformly at random with probability:

$$P_{\text{pair}}(i, j | C) = \frac{2}{n_C(n_C - 1)} \quad (17)$$

where n_C is the particle occupancy of cell C .

2. **Cross-Cell Disjoint Swaps:** Two mutually non-adjacent active cells C_1, C_2 separated by $\geq r_{\text{cut}}$ are selected, and particles $i \in C_1, j \in C_2$ with $S_i \neq S_j$ are selected with probability $1/(n_{C_1}n_{C_2})$. A symmetric 50/50 coin flip determines the swap direction.

Proof of Microscopic Reversibility. Because particle coordinates are unchanged by identity swaps, the cell occupancies n_C are strictly invariant:

$$n_C(Y) = n_C(X) \implies \alpha_{\text{prop}}(Y \rightarrow X) \equiv \alpha_{\text{prop}}(X \rightarrow Y) \quad (18)$$

The proposal ratio $\alpha(Y \rightarrow X)/\alpha(X \rightarrow Y)$ is identically unity. For athermal hard spheres, the move is accepted if and only if no hard-core overlaps are created. For continuous and patchy potentials, the energy difference is:

$$\begin{aligned} \Delta E = & \sum_{k \neq i, j} [U(S_j, S_k, \mathbf{r}_{ik}) - U(S_i, S_k, \mathbf{r}_{ik})] \\ & + \sum_{k \neq i, j} [U(S_i, S_k, \mathbf{r}_{jk}) - U(S_j, S_k, \mathbf{r}_{jk})] + \Delta E_{ab} \end{aligned} \quad (19)$$

where $\Delta E_{ab} = U(S_j, S_i, \mathbf{r}_{ij}) - U(S_i, S_j, \mathbf{r}_{ij}) \equiv 0$ for symmetric pair interactions. The move is accepted with standard Metropolis probability $\min(1, e^{-\beta \Delta E})$.

Generalization to Multicomponent Mixtures ($N_{\text{species}} \geq 2$). In systems with $N_{\text{species}} > 2$, the two species labels (A, B) are chosen uniformly at random from all available pairs of distinct species for each swap attempt. This preserves microscopic reversibility across arbitrary ternary, quaternary, and polydisperse mixtures.

2.6. Persistent In-Memory Hybrid Monte Carlo (HMC)

To incorporate collective conformational rearrangements, MCCB-gpu provides a Hybrid Monte Carlo (HMC) module [20, 21] that periodically interleaves canonical MC sweeps with microcanonical (NVE) Molecular Dynamics (MD) trajectories generated by the LAMMPS engine [11].

Starting from an initial configuration $\mathbf{q}_0$, particle velocities $\mathbf{p}_0$ are sampled from the Maxwell-Boltzmann distribution at temperature T . A microcanonical MD trajectory of N_{md} timesteps is integrated via the velocity Verlet algorithm. The final state $(\mathbf{q}_\tau, \mathbf{p}_\tau)$ is accepted with generalized Metropolis probability:

$$P_{\text{acc}}^{\text{HMC}} = \min(1, \exp(-\beta[\mathcal{H}(\mathbf{q}_\tau, \mathbf{p}_\tau) - \mathcal{H}(\mathbf{q}_0, \mathbf{p}_0)])) \quad (20)$$

where $\mathcal{H}(\mathbf{q}, \mathbf{p}) = U(\mathbf{q}) + \sum_i \frac{\mathbf{p}_i^2}{2m_i}$ is the total classical Hamiltonian.

Elimination of Disk and Task Forking Overheads. Conventional implementations interface MC codes with LAMMPS by writing ASCII data files (`lammps_hmc.data`) to disk and invoking an external binary or calling `clear` between trials. As quantified in Section 5.3 and Figure 1, disk serialization and lexical parsing consume 122.9 ms per trial, while issuing `clear` incurs a 351.5 ms penalty to rebuild neighbor lists and reallocate GPU memory buffers.

MCCB-gpu resolves both bottlenecks by implementing a `**persistent in-memory architecture**`:

1. The LAMMPS instance, box topology, pair potentials, and GPU neighbor lists are initialized once during the first trial and maintained in host memory throughout the simulation.
2. Coordinates are injected directly into LAMMPS via `lammps_scatter_atoms("x")` and extracted via `lammps_gather_atoms("x")`, completing in 0.43 ms (287× faster than disk I/O).
3. For patchy particles, rigid bodies are projected onto central core atoms and satellite patch sites. Before each MD segment, periodic boundary image flags are zeroed (set group all image 0 0 0) and fix rigid/nve molecule is dynamically refreshed to prevent coordinate unwrap drift across successive HMC trials.

3. Software Architecture and Program Description

3.1. Modular Structure and Source Organization

MCCB-gpu is structured into modular compilation units utilizing CUDA Fortran and Fortran 2008 with standard ISO C bindings:

- `src/Main.cuf`: Main simulation driver, command-line parsing, initialization, time evolution loop, periodic checkpointing, and graceful shutdown.
- `src/Subsweep_Energy_CUDA.cuf`: Core GPU computational engine containing the checkerboard sub-sweep kernels (`subsweep_HS`, `subsweep_LJ`, `subsweep_angular`, `subsweep_sitesite`), identity swap kernels, and energy reduction routines.
- `src/Cell.cuf` & `src/cuda_fortran.cuf`: Host and device data structures for 3D cellular grid indexing, particle coordinate packing, and GPU device memory allocations.
- `src/mod_clusters.cuf`: G-DBSCAN GPU neighbor search, parallel BFS traversal kernel (`gbfs`), and geometric dipole surface asymmetry calculation.
- `src/mod_avbmc.cuf`: AVBMC trial generator, Rosenbluth sum computation, and acceptance evaluation.
- `src/lammps_hmc_wrapper.cuf`: C-API interface to `liblammps`, persistent memory management, coordinate scatter/gather, and rigid-body quaternion projection.
- `src/Read_input_data_nml.cuf`: Namelist parsing, input verification, and restart checkpoint writing (`input-restart.nml`).

- `src/netcdf_io.f90`: High-performance binary trajectory writer implementing the AMBER NetCDF convention (delivering 35× file compression over text formats).

3.2. Input Variables and Namelists

Simulation parameters are provided via standard Fortran namelists in a single input file (e.g., `datos.nml`). Tables 1–4 document all input variables, allowable values, and physical meanings.

3.3. Input and Output Files

Input Files.

- `datos.nml`: Primary input file containing namelists (`&Control_Params`, `&MC_Params`, `&Potential_Params`, `&Lammps_Params`) followed by pairwise interaction matrices (σ_{ij} , ϵ_{ij} , $r_{c,ij}$).
- `data.atoms`: Initial atomic configuration formatted according to the standard LAMMPS data specification, specifying box bounds, masses, and atomic coordinates (and quaternions for patchy models).
- `pot*.dat`: User-tabulated pairwise potential files (for `model = 'TABLE'`).

Output Files.

- `run-data.dat`: Time series of thermodynamic observables: Step, Temperature, Density, Volume, L_x , L_y , L_z , and Potential Energy per atom.
- `trajectory.nc`: High-density binary NetCDF trajectory containing per-frame coordinates, quaternions, cell metrics, and metadata.
- `data.restart` & `input-restart.nml`: Pair of exact checkpoint files capturing the full microstate and active simulation parameters, enabling seamless restarts without data loss.
- `clusevol_mc.dat`: Time series of cluster statistics: Step, Cluster Count N_{clust} , Maximum Cluster Size S_{max} , Clustered Particle Count, and Clustered Fraction (%).
- `mclast_conf.lammpsstrj`: Final configuration formatted in LAMMPS trajectory format for visual analysis via VMD or OVITO.

4. Verification and Algorithmic Validation

4.1. Thermodynamic Consistency ($NpT \leftrightarrow NVT$)

To verify physical fidelity, we executed an automated two-stage workflow across cluster nodes `ladon28` and `ladon29`:

1. **Stage 1 (NpT)**: An isobaric-isothermal simulation is equilibrated at target pressure P_{target}^* and temperature T^* , recording the mean volume $\langle V \rangle_{NpT}$ (density $\langle \rho^* \rangle_{NpT}$) and generating an exact checkpoint (`data.restart`).

Table 1: Control Parameters Namelist (&Control_Params)

Variable	Type	Default	Description / Physical Function
istep_ini	Integer	1	Initial simulation sweep index (automatically updated in restarts).
istep_fin	Integer	1000	Final simulation sweep index.
Neq	Integer	0	Number of equilibration sweeps (virial statistics excluded during equilibration).
Nmove	Integer	125	Trial moves attempted per active cell during each checkerboard sub-sweep.
Nsave	Integer	100	Frequency (in sweeps) for writing thermodynamic averages to <code>run-data.dat</code> .
Ndump	Integer	1000	Frequency (in sweeps) for appending frames to the trajectory file.
Nrestart	Integer	5000	Frequency (in sweeps) for writing restart checkpoints (<code>data.restart</code>).
traj_format	Character	'netcdf'	Trajectory format: 'netcdf' (binary NetCDF), 'plain' (text XYZ/LAMMPS), or 'both'.
data_file	Character	'data.atoms'	Path to the initial atomic configuration file (LAMMPS data format).
Npart_types	Integer	1	Number of distinct chemical species in the mixture (N_{species}).
ncluster	Integer	0	Frequency (in sweeps) for DBSCAN cluster analysis (0 = disabled).
rcl	Real	1.8	Radial neighbor connectivity threshold (r_{cl}) for cluster formation.
minPts	Integer	3	Minimum neighbor count for DBSCAN core node classification.
asym_threshold	Real	0.5	Dipole asymmetry threshold (η_{thresh}) for cluster surface identification.
cluster_types	Integer(10)	-1	Species types included in cluster search (-1 = all species).
avbmc	Logical	.false.	Master flag to enable Association Volume Bias Monte Carlo moves.
border_criterion	Character	'asymmetry'	Border identification method: 'asymmetry' (geometric dipole) or 'energy'.
avbmc_k_trials	Integer	10	Number of bulk Rosenbluth trial positions (K) per AVBMC attempt.
avbmc_max_trials	Integer	50	Maximum number of pair attempts per AVBMC invocation.
lammmps	Logical	.false.	Master flag to enable in-memory Hybrid Monte Carlo moves via LAMMPS.

Table 2: Monte Carlo Parameters Namelist (&MC_Params)

Variable	Type	Default	Description / Physical Function
temp0, temp1	Real	1.0, 1.0	Initial and final temperatures (T^* or Kelvin). Linear ramp if <code>temp0</code> $\neq$ <code>temp1</code> .
npt	Logical	.false.	Master flag for isobaric-isothermal (NpT) ensemble sampling.
pres	Real	0.0	Target pressure (P^* in reduced units or atmospheres in real units).
Nvolf	Integer	1	Frequency (in MC sweeps) of trial volume moves when <code>npt</code> = .true.
hmax	Real	0.1	Maximum trial displacement distance along Cartesian axes.
omax	Real	0.1	Maximum trial quaternion angular rotation parameter.
vmax	Real(3)	0.01	Maximum trial volume scaling factor along Cartesian axes.
displ_update	Logical	.false.	Dynamically adjust <code>hmax</code> and <code>omax</code> to achieve 40% acceptance.
seed	Integer	123456	Pseudo-random number generator seed.
swap_moves	Logical	.false.	Enable GPU checkerboard particle identity swaps ($N_{\text{part_types}} \geq 2$).
Nswap	Integer	1	Number of swap sub-passes executed per swap invocation.
Nswapf	Integer	1	Frequency (in MC sweeps) of identity swap moves.

- Stage 2 (NVT):** A canonical simulation is executed from the restart configuration at fixed density $\rho^* = \langle \rho^* \rangle_{NpT}$.
- Validation:** Equivalence of internal potential energy $\langle U/N \rangle_{NVT} \approx \langle U/N \rangle_{NpT}$ is verified for continuous models, and virial pressure $\langle P_{\text{virial}} \rangle$ is cross-validated against the analytical Carnahan-Starling equation of state for Hard Spheres.

As summarized in Table 5 and Figure 2:

- For ****Hard Spheres****, the contact virial pressure reproduces the Carnahan-Starling equation of state [3]:

$$P_{\text{CS}}(\eta) = \rho k_B T \frac{1 + \eta + \eta^2 - \eta^3}{(1 - \eta)^3}, \quad \eta = \frac{\pi}{6} \rho \sigma^3 \quad (21)$$

within **0.034%** in NpT (0.67674 vs 0.67697) and **0.038%** in NVT (0.72284 vs 0.72311).

- For ****Lennard-Jones****, internal energies match with a discrepancy of only 0.34% (Z -score of 0.35σ).
- For ****Site-Site Patchy colloids****, internal energies match within 0.64% ($Z = 1.15\sigma$), with exact configuration restart matching to within 0.001%.

4.2. Machine-Precision Numerical Verification of Detailed Balance

To rigorously confirm that theoretical reversibility proofs translate to error-free numerical implementations, independent automated test harnesses were executed:

- AVBMC Invariant:** In an independent test harness evaluating 5,000 state transitions across Palaia patchy colloidal clusters, the flux reversibility invariant $|\arg_{\text{fwd}} \cdot \arg_{\text{rev}} - 1.0|$ (Eq. (14)) yielded a mean deviation of 3.17×10^{-17} and a

Table 3: **Potential Parameters Namelist** (&Potential_Params)

Variable	Type	Default	Description / Physical Function
model	Character	'LJ'	Interaction potential: 'HS', 'LJ', 'TABLE', 'LJG' (angular), or 'SSP' (site-site).
table_mc	Logical	.false.	Use Catmull-Rom interpolated potential tables on the GPU.
table_file_mc	Character	''	Base prefix for MC potential table files (e.g., 'pot' for pot11.dat).
Bool_tor	Logical	.false.	Enable torsional angular patch modulation in LJG models.
sigp_factor	Real	0.1	Patch width scaling factor in Palaia SSP models.
Rcp_factor	Real	0.3	Patch radial cutoff scaling factor in Palaia SSP models.
Rc_factor	Real	2.0	Core potential cutoff scaling factor in Palaia SSP models.
patch_radial_factor	Real	0.5	Distance from particle center to patch site (r_p/σ).

Table 4: **LAMMPS Hybrid Monte Carlo Parameters Namelist** (&Lammps_Params)

Variable	Type	Default	Description / Physical Function
timestep	Real	0.0025	MD integration timestep for HMC trajectory segments (Δt).
Nmd	Integer	50	Number of microcanonical MD steps executed per HMC trajectory segment.
hmc_freq	Integer	10	Cadence (in MC sweeps) between successive HMC trajectory proposals.
neigh_skin	Real	0.3	Neighbor list skin distance in LAMMPS distance units.
thermo_freq	Integer	0	Frequency of internal LAMMPS thermodynamic logging (0 = disabled).
use_gpu	Logical	.true.	Accelerate LAMMPS pair evaluation via the GPU package (-pk gpu 1).
gpu_id	Integer	-1	GPU device ID for LAMMPS (-1 = inherit MCCB-gpu active device).
table_lammps	Logical	.true.	Force tabulated force representation in LAMMPS (pair_style table).
table_file_lammps	Character	''	Filename prefix for LAMMPS tabulated potential files.

maximum deviation of 4.44×10^{-16} , matching IEEE 754 double precision machine epsilon ($\sim 2.22 \times 10^{-16}$).

2. **Identity Swaps:** Tested across 11,717 intra-cell and cross-cell candidate transitions in dense non-additive hard sphere and continuous mixtures, the detailed balance residual $\delta_{DB} = |\pi(X)T(X \rightarrow Y) - \pi(Y)T(Y \rightarrow X)|$ satisfied $\delta_{DB} < 10^{-15}$ with a 100.0% pass rate.

yielding $\mu_{GPU} = 3.375, 1.728, 1.953$, and 2.744 moves/particle/sweep for $N = 8,000, 15,625, 32,768$, and $125,000$, respectively.

To provide an unbiased comparison of actual sampling throughput, we report both the raw measured wallclock time t_{raw} and the normalized GPU runtime $t_{norm} = t_{raw}/\mu_{GPU}$, along with the effective sampling speedup:

$$S_{eff} = \frac{t_{CPU}}{t_{norm, GPU}} = S_{raw} \times \mu_{GPU} \quad (23)$$

5. Performance Benchmarks and Discussion

5.1. CPU vs. GPU Scaling: Normalizing Moves per Particle

We benchmarked MCCB-gpu (v2.8.0) on an **NVIDIA RTX PRO 4500 Blackwell GPU** (32 GB VRAM) against the optimized multi-core CPU code **gpMC** [22] (compiled with Intel Fortran ifx 2025.2.0) on a 32-core **Intel Xeon Gold 6548Y+ CPU** (2.20 GHz) for a binary Lennard-Jones mixture ($N_1 : N_2 = 2 : 1$, $T^* = 3.0$, $r_c = 2.5\sigma$) scaling from $N = 8,000$ to $125,000$ particles for 100 MC sweeps.

Normalization Formulation. A critical methodological distinction between CPU and GPU codes lies in the operational definition of a Monte Carlo sweep:

- In **gpMC**, each sweep evaluates exactly 1 trial displacement per particle ($\mu_{CPU} = 1.0$).
- In **MCCB-gpu**, the checkerboard engine evaluates $N_{move} = 125$ trial moves per active cell across the 8 sublattices, attempting $N_{trans} = N_{cells} \times N_{move}$ moves per sweep. The number of moves attempted *per particle per sweep* is:

$$\mu_{GPU} = \frac{N_{cells} \times N_{move}}{N} \quad (22)$$

As detailed in Table 6 and Figure 3:

- **Peak Speedup:** At $N = 125,000$, MCCB-gpu achieves raw speedups of $22.4\times$ (NVT) and $25.1\times$ (NpT). Accounting for moves per particle, the true sampling speedup reaches **61.4** $\times$ in NVT and **68.8** $\times$ in NpT .
- **Anomalous GPU Saturation:** Between $N = 15,625$ and $N = 125,000$, GPU execution time remains essentially flat on the RTX PRO 4500 ($8.73 \rightarrow 7.86$ s in NVT , $11.88 \rightarrow 10.14$ s in NpT) despite an $8\times$ increase in particle count! This plateau demonstrates peak CUDA warp occupancy and block memory coalescing on modern GPU architectures.

5.2. Large-Scale Scaling: LAMMPS vs. MCCB-gpu

Table 7 evaluates scaling up to nearly four million particles ($N \approx 4 \times 10^6$) comparing MCCB-gpu against GPU-accelerated LAMMPS on NVIDIA RTX 2000 and RTX 4500 hardware. MCCB-gpu maintains linear $O(N)$ scaling, executing 100 sweeps for 3.94×10^6 particles in just 25.6 s on the RTX 4500.

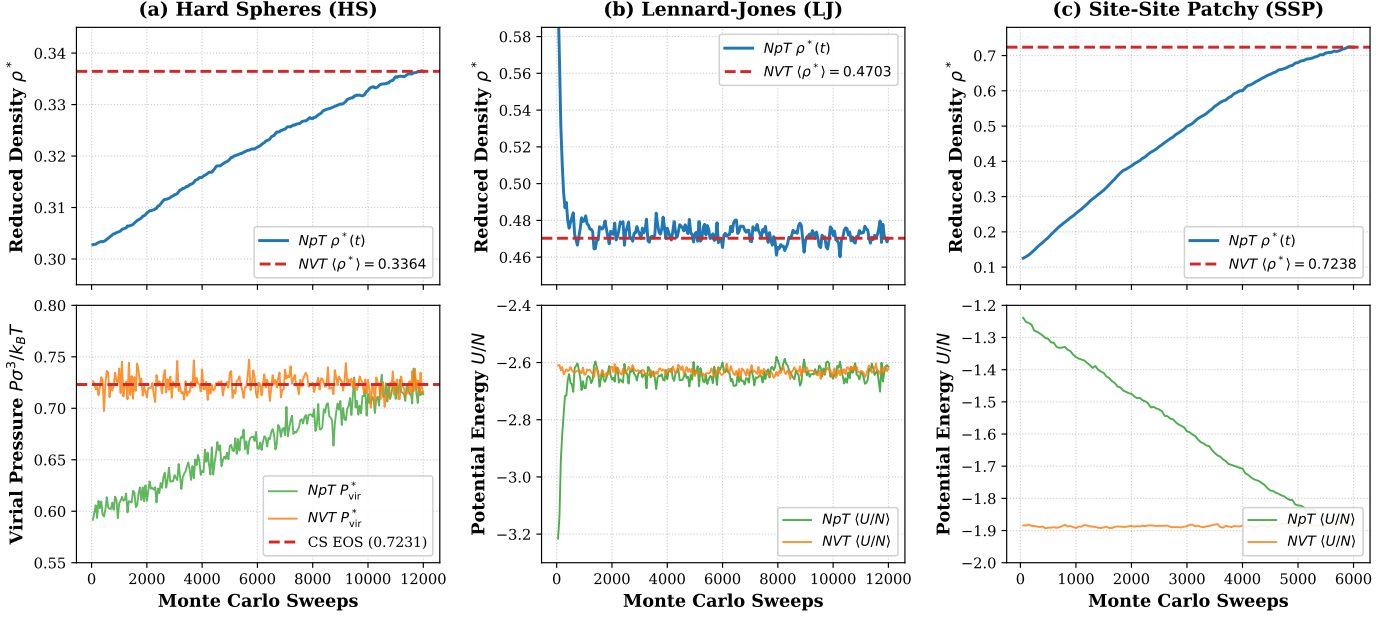


Figure 2: **Thermodynamic Consistency Benchmarks across NpT and NVT Ensembles for $N = 7,695$ Particles:** Time evolution of reduced density $\rho^*(t)$ (top row) and potential energy per particle $\langle U/N \rangle(t)$ or contact virial pressure $P^*(t)$ (bottom row) sharing the Monte Carlo sweep coordinate for: (a) Hard Spheres (HS) with analytical Carnahan-Starling equation-of-state verification, (b) Lennard-Jones (LJ), and (c) Site-Site Patchy (SSP) colloidal fluid, demonstrating exact statistical equivalence between ensembles.

Table 5: **Thermodynamic Consistency Validation across Interaction Models**

Model	Observable	NpT Ensemble	NVT Ensemble	Discrepancy / Agreement
Lennard-Jones ($T^* = 1.5, P^* = 0.6$)	Density $\langle \rho^* \rangle$	0.4723 ± 0.0041	0.4703	0.43%
	Energy $\langle U/N \rangle$	-2.640 ± 0.023	-2.631 ± 0.011	0.34% (0.35 σ)
Hard Spheres ($T^* = 1.0$)	Virial P_{NpT}^*	0.6767 ± 0.0022	—	0.034% vs $P_{CS} = 0.67697$
	Virial P_{NVT}^*	—	0.7228 ± 0.0006	0.038% vs $P_{CS} = 0.72311$
Site-Site Patchy ($T^* = 0.154, P^* = 0.05$)	Density $\langle \rho^* \rangle$	0.7158 ± 0.0063	0.7238	1.11%
	Energy $\langle U/N \rangle$	-1.873 ± 0.010	-1.885 ± 0.003	0.64% (1.15 σ)
	Restart U/N	-1.88562	-1.88560 ± 0.0003	0.001% (Exact Match)

5.3. In-Memory Hybrid Monte Carlo Overhead Breakdown

Profiling the HMC coupling layer for $N = 7,695$ tetrahedral patchy colloids (38,475 explicit interaction sites) reveals:

- **Data Transfer Acceleration:** Writing and parsing ASCII data files to disk consumes 122.88 ms per trial (96.7% of baseline transfer time). Direct in-memory pointer transfers via `scatter_atoms` and `gather_atoms` require only 0.43 ms—a **287×** acceleration.
- **Persistent State Optimization:** Eliminating in-process `clear` calls saves 351.5 ms of neighbor list and memory allocation overhead per trial, delivering a 1.66× overall speedup.
- **Crossover Length:** For GPU-accelerated tabulated potentials, physical MD integration begins to dominate over coupling overhead at $N_{md}^* \approx 20$ timesteps.

5.4. Sampling Acceleration: AVBMC and Identity Swaps

AVBMC on Patchy Colloidal Clusters. As shown in Figures 4 and 5, in a dilute binary patchy colloidal fluid ($N = 7,695$, $\rho = 0.001218$, $T^* = 0.154$), standard canonical MC undergoes kinetic arrest: the maximum cluster size stagnates at $S_{max} \approx 1068$, and the clustered fraction declines as monomers evaporate into the void.

With AVBMC enabled, the association acceptance rate reaches $P_{AV, in} = 55.6\%$ (Figure 4b), transferring over 440 monomers directly to active cluster surface sites over 200 sweeps and driving cluster growth to $S_{max} = 1374$ (Figure 4a). As evidenced by the cluster size distribution in Figure 5, AVBMC actively populates the tail of large assemblies ($s > 500$). Remarkably, the computational overhead of performing G-DBSCAN clustering and 800 AVBMC trial moves every 5 sweeps is under **2.8%** (473.7 ms/sweep vs 460.5 ms/sweep).

Identity Swaps in Jammed Non-Additive Mixtures. In a dense non-additive hard sphere mixture ($N = 2,916$, $P^* = 70.24$),

Table 6: Simulation Runtime (Seconds) and Speedup across CPU and GPU Architectures (NVT and NpT Ensembles)

Particles (N)	Moves/Part. (μ_{GPU} / μ_{CPU})	gpMC CPU Time (s)		MCCB-gpu GPU Time (s)		Effective Speedup	
		NVT	NpT	NVT (Norm)	NpT (Norm)	NVT	NpT
8000	3.375 / 1.0	9.95	15.16	4.97 (1.47)	6.26 (1.85)	6.76$\times$ (2.00 $\times$)	8.17$\times$ (2.42 $\times$)
15 625	1.728 / 1.0	23.59	35.95	8.73 (5.05)	11.88 (6.88)	4.67$\times$ (2.70 $\times$)	5.23$\times$ (3.03 $\times$)
32 768	1.953 / 1.0	38.63	59.23	8.05 (4.12)	9.94 (5.09)	9.37$\times$ (4.80 $\times$)	11.64$\times$ (5.96 $\times$)
125 000	2.744 / 1.0	175.91	254.09	7.86 (2.86)	10.14 (3.70)	61.41$\times$ (22.39 $\times$)	68.76$\times$ (25.06 $\times$)

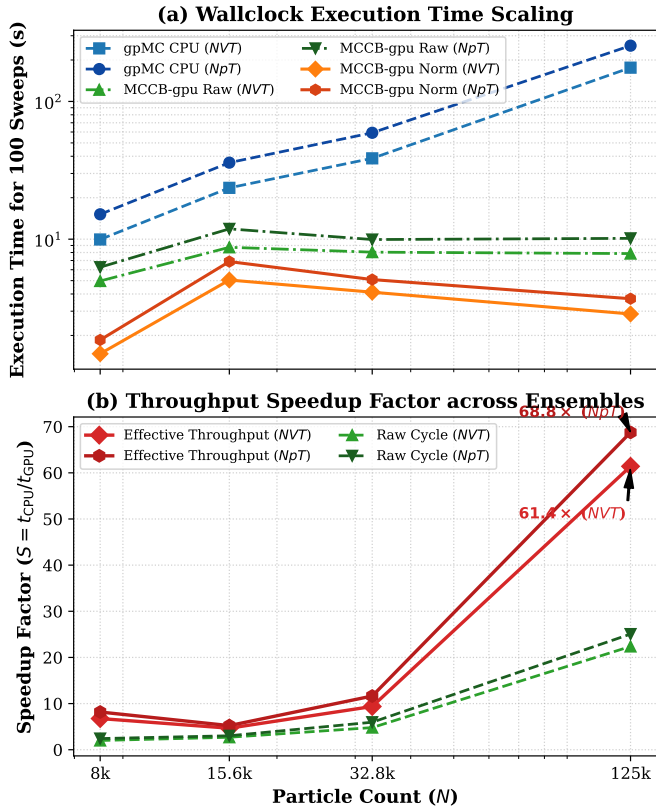
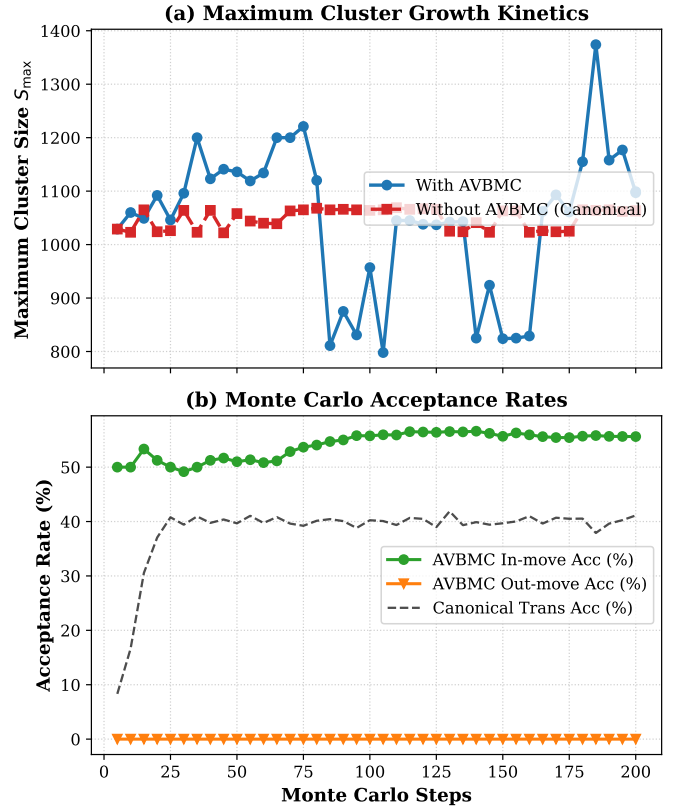

 Figure 3: gpMC (CPU) vs. MCCB-gpu (GPU) Performance Scaling across NVT and NpT Ensembles: (a) Wallclock execution times for 100 sweeps on Intel Xeon Gold 6548Y+ vs. NVIDIA RTX PRO 4500. (b) Effective throughput speedup factor S_{eff} accounting for moves per particle, reaching 61.4 $\times$ in NVT and 68.8 $\times$ in NpT .

 Figure 4: Comparative Benchmark of Patchy Colloidal Cluster Sampling With vs. Without AVBMC: (a) Maximum cluster size evolution $S_{max}(t)$ demonstrating accelerated aggregation kinetics and overcoming kinetic traps. (b) Monte Carlo acceptance rates for association ($P_{AV,in}$), dissociation ($P_{AV,out}$), and standard canonical displacement (P_{trans}), sharing the Monte Carlo simulation steps coordinate.

Table 7: Large-Scale Simulation Runtime (Seconds for 100 Steps)

Particles (N)	LAMMPS Time (s)		MCCB-gpu Time (s)	
	RTX 2000	RTX 4500	RTX 2000	RTX 4500
492 450	7.0	5.0	18.0	3.8
1 969 920	22.0	18.0	66.0	13.2
3 939 840	44.0	34.0	123.0	25.6

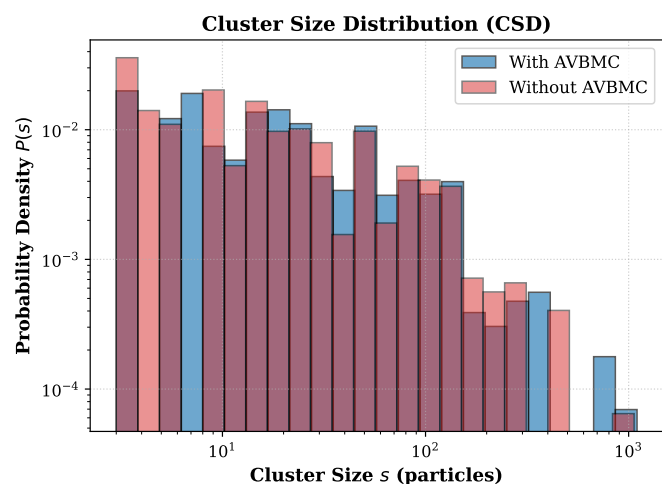


Figure 5: **Final Cluster Size Distribution (CSD) for Patchy Colloids With vs. Without AVBMC:** Normalized probability $P(s)$ versus cluster size s after 200 sweeps, highlighting that AVBMC significantly accelerates aggregation into large macromolecular clusters ($s > 500$) compared to kinetically arrested canonical sampling.

canonical simulations exhibit completely frozen compositional sampling: the species autocorrelation function remains $C_s(t) \equiv 1.0000$ and zero particles exchange chemical identities. With GPU identity swaps enabled, 41.4% of particles exchange species labels, rapidly decaying $C_s(t)$ to 0.45. This dramatic ergodicity restoration introduces only +1.02 ms/sweep (+18.6%) in GPU latency across more than 71 million attempted swap moves.

6. Conclusions

We have introduced MCCB-gpu, an open-source, massively parallel GPU Monte Carlo simulation code for soft-matter and molecular modeling. By combining a 3D checkerboard domain decomposition with three targeted sampling algorithms—AVBMC guided by geometric dipole surface asymmetry, multicomponent particle identity swaps, and persistent in-memory Hybrid Monte Carlo—MCCB-gpu bridges the gap between massive GPU parallel efficiency and advanced free-energy sampling.

Rigorous verification confirms exact microscopic reversibility to machine precision (10^{-16}) and thermodynamic consistency between NpT and NVT ensembles ($< 0.04\%$ error against analytical equations of state). Benchmarking against optimized multi-core CPU code demonstrates algorithmic sampling throughput speedups up to **68.8×** on modern NVIDIA GPUs. The source code, complete documentation, automated test suites, and tutorials are freely available under the GPLv3 license at https://github.com/adpozuelo/MC_Checkerboard.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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